

Climate Change and Its Impact on Children and Schools

Report by- **Swaraj kumar**

Climate change, especially heat extremes, is increasingly affecting the health and education of children in India and the world. In 2024, more than 242 million children worldwide stayed home from school due to heat extremes. India and South Asia had the largest impacts, with more than 118 million children affected within just a month. These events not only lead to schooldays lost but also long-term damage to learning, emotions, and development. Indian schools, particularly government schools, face problems like poor infrastructure, poor sanitation, and no system to deal with heat. But there are good policy and education ideas starting to come through to deal with this new problem. Based on the data from the publications such as Amar Ujala, The Times of India, NDTV and Dainik Jagran.

1. Heat Stress and School Closures

- India's Vulnerability

54.8 million Indian school children had problems with school because of heatwaves in 2024 ([Hindustan Times](<https://www.hindustantimes.com>))

UNICEF reported that 242 million children in the world did not attend school due to severe weather. 128 million of them were located in South Asia.

- Loss in Learning and School Skills

A 1 °C rise in temperature can reduce students' performance by as much as 1%. Warm years can result in 1–1.5 years of schooling loss ([Amar Ujala](<https://results.amarujala.com>)).

In India, there are short or no school days when indoor temperatures are more than 45 °C. This amounts to approximately a~10% yearly loss of learning.

- Without Infrastructure

4% of Indian government schools lack clean drinking water, and 6% lack toilets, which disproportionately affect girls' attendance.

Tin-roof classrooms with no fans or insulation become impossible to live in during summer.

2. Hydration, Well-being & Health

- Health Risks

Heat exposure can cause dehydration, heatstroke, and tiredness in school or on the way.

Girls are more affected by inadequate management of menstrual hygiene in hot, under-resourced schools.

- Effects on Thinking and Mental Wellbeing

Heat deteriorates concentration, memory, and attention spans.

Increased temperatures and humidity are associated with greater depression and anxiety in students.

3. Education Interruptions & Learning Loss

- School closures, absences, and dropouts

Climate activities result in numerous school closures, delayed exams, and premature summer breaks, particularly in northern and central India ([\[Hindustan Times\]\(https://www.hindustantimes.com\)](https://www.hindustantimes.com)).

75% School closures due to severe weather occur in low- and middle-income countries, like India ([\[Amar Ujala\]\(https://results.amarujala.com\)](https://results.amarujala.com)).

- Long-Term Developmental Effects

Early exposure to climate shocks reduces numeracy, vocabulary, and non-cognitive skills by adolescence.

India ranks 26th among 163 nations on the UNICEF Children's Climate Risk Index. Stunting will rise by 35% by 2050.

4. Policy Response and Education

- Awareness and Curriculum Programs

Delhi government is introducing climate education modules in 10 pilot girls' schools (2025–26).

200,000 students will be educated by eco-clubs with assistance from ([TOI](https://timesofindia.indiatimes.com)).

- Engagement Through Storytelling & Campaigns

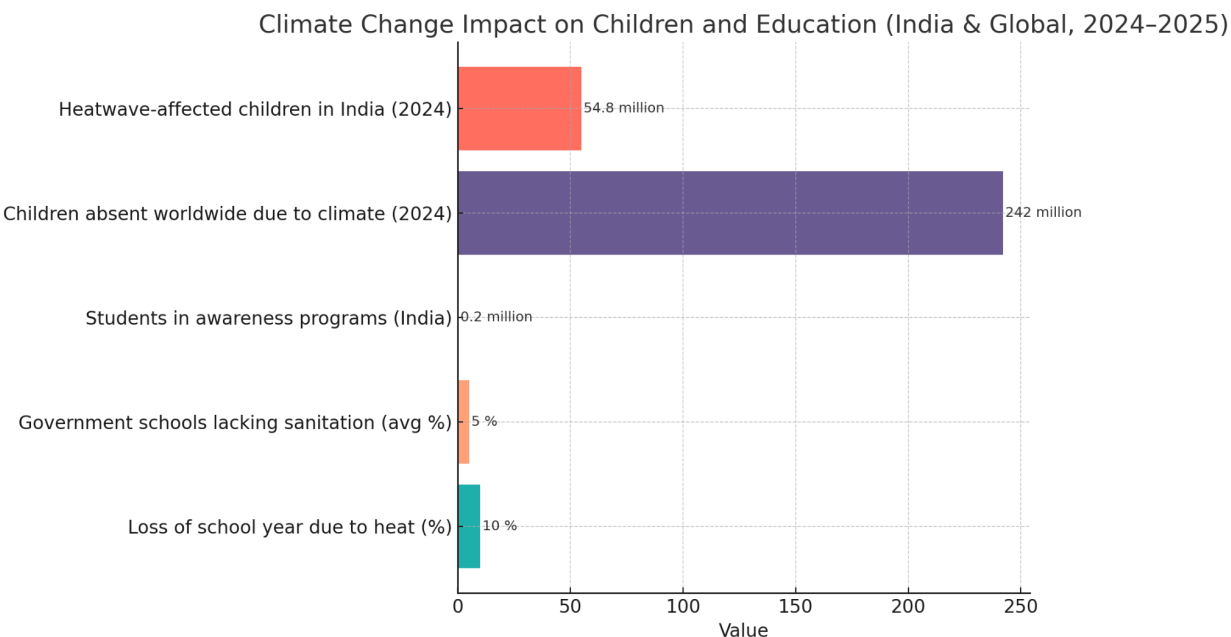
More than 580 students from UP's KGBV schools took part in climate-themed storytelling competitions ([TOI](https://timesofindia.indiatimes.com)).

- Infrastructure Innovations

In Meghalaya, a solar-powered classroom was inaugurated. It marries climate resilience with digital learning ([TOI](https://timesofindia.indiatimes.com)).

Delhi's "MY Bharat" initiative engages students in monthly climate missions—plantations, waste segregation, clean air drives ([TOI](https://timesofindia.indiatimes.com)).

5. A concise Chart



6. Suggestions

To meet this vital challenge, the following steps are suggested:

1. Renovate buildings in schools to suit the changing climate, improved ventilation, air conditioning, shading outside, solar panels, and greenery.
2. Ensure safe water, proper sanitation, and hygiene (WASH) particularly for adolescent girls.
3. Alter school schedules to accommodate climate trends morning school, blended learning strategies, or taking breaks during specific periods.
4. Include climate change in lessons helping students learn about the environment and gain control.
5. Encourage mental and physical well-being hydration breaks, first aid, and heat illness prevention school initiatives.
6. Scale up youth climate action eco-clubs, climate storytelling competitions, and youth-led campaigns can drive ownership and awareness.
7. Establish strong national and local data systems to track climate-related education challenges and measure solutions in real-time.

Last Idea Not only is climate change impacting education, but education can also be used to correct it. If we invest in climate smart education systems now, we can produce a generation of resilient problem solvers and leaders who are equipped to take us on the path to a sustainable future.

7. Conclusion

Climate change is not something in the future—it's already impacting education, health, and development for millions of children today, particularly in vulnerable locations like South Asia. In India, recurring heatwaves and inadequate infrastructure are pushing children, particularly girls, out of school and out of the opportunity to thrive. If we delay, these issues will have lasting economic and social costs.